

1) Find out the history of Towers of Hanoi

A Tower of Hanoi was invented in 1883 by French mathematician Edouard Lucas.

→ The puzzle consists of

(1) Three rods

(2) A no. of disks of diff sizes stacked on one rod.

(3) The goal is to move the entire stack to another rod following rules

(i) Move only one disk at a time

(ii) Only the top disk of a stack may be moved

(iii) Larger disk is not allowed to place on smaller disk.

→ It became an important example mathematics and Computer Science because it demonstrates

(1) Recursive problem solving

(2) Mathematical Induction

(3) Algorithm design

→ Lucas attached a fictional story i.e Tower of Brahma

→ In temple, 64 disks are moved by monks b/w 3 diamond pegs.

→ When they complete the task, the world will end.

2) DSA observations in Surroundings

1) Alarms on Mobile

→ Multiple alarms ring one after another.

2) Phone Contacts

→ Contacts are stored in an organised manner making searching faster

3) College Bus

→ Students boards one after another, those who get first get the seat first

4) Classroom Attendance:-

The attendance register stores students in a sequential list.

5) College Canteens:-

Students wait in line, student in queue is served first.

6) Web Browser:-

Back button returns to the previously visited page.

7) Google Maps:-

Maps calculate shortest and fastest route to destination.

8) Whatsapp Messages:-

Messages are display in the order they are sent.